Classwork

1. $z = wx + b$

$= (2 \times 3) + -1$

$= 5$

2. $z = wx + b$ $\qquad y = \sigma(z) = \frac{1}{1+e^{-z}}$

$= 1.5 \cdot 1 - 0.5$ $\qquad = \frac{1}{1+e^{-1}}$

$= 1$ $\qquad = 0.731$

3. $z = wx + b = 3 \cdot -2 + 1 = -5$ $\qquad a = \max(0, -5) = 0$

4. $\frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}(4-1)^2 = \frac{9}{2} = 4.5$

5. $\frac{\partial L}{\partial y} = \hat{y} - y = 6 - 10 = -4$

6. $\hat{y} = 1 \times 2 + 0 = 2$ $\qquad \hat{y} - y = 2 - 5 = -3$

$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot x = -3 \cdot 2 = -6$

$\frac{\partial L}{\partial b} = (\hat{y} - y) \cdot 1 = -3 \cdot 1 = -3$

7. $w_{new} = w - \eta \frac{\partial L}{\partial w} = 1 - (0.1 \times -6) = 1 + 0.6 = 1.6$

$b_{new} = b - \eta \frac{\partial L}{\partial b} = 0 - (0.1 \cdot -3) = 0 + 0.3 = 0.3$

$w = 1.6$, $b = 0.3$

Scanned with CamScanner

8.

Step 1:

$\hat{y} = 0$

$\frac{\partial L}{\partial w} = (\hat{y} - y)x = -2$, $\frac{\partial L}{\partial b} = (\hat{y} - y) = -2$

$w_1 = 0 - 0.5 \cdot -2 = 1$

$b_1 = 0 - 0.5 \cdot -2 = 1$

Step2:

$\hat{y} = 2$

$\frac{\partial L}{\partial w} = (2-2)1 = 0$ $\frac{\partial L}{\partial b} = 0$

Thus, we have no update

$w_2 = 1$

$b_2 = 1$

Scanned with CamScanner